Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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Professional Summary

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

Core Competencies

Avionics (MIL-STD-1553 / ARINC 429, Avionics integration) · Scientific instrumentation (Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems) · Embedded systems (Embedded C, ESP32 / STM32 microcontrollers) · Computer vision (Image super-resolution, Object detection) · Drones & UAV systems (UAV payload integration) · Edge AI (NVIDIA Jetson deployment) · AI / machine learning (Deep learning, PyTorch) · Procurement & administration (Technical procurement (GeM, GFR)) · IoT (MQTT and Node-RED, InfluxDB and Grafana) · Model optimisation (TensorRT and ONNX, Pruning and knowledge distillation) · Software & tools (Python, Docker and Linux) · Technical leadership (Team leadership)

Professional Experience

Scientist E Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

Senior Design Engineer Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

Graduate Engineer Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

Key Projects

Edge super-resolution pipeline for UAV imagery *(Ongoing)* Sep 2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Fictional sample profile — not a real person

Education

Ph.D., Electrical and Electronics Engineering 2021 – ongoing
Northfield Institute of Science and Technology
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics 2010
Coastal Institute of Aeronautical Engineering, Chennai
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering 2004
Northfield Institute of Science and Technology, Northfield University, Pune
Grade: First Class with Distinction

Publications

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

Leadership & Responsibilities

Supervisor, M.Tech. interns in applied machine learning 2022 – Present
Kestrel Atmospheric Instrumentation Laboratory
Supervise summer and dissertation interns on edge-AI projects.

Chair, Technical Evaluation Committee for radiosonde procurement Apr 2023 – Feb 2024
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

Awards & Honours

Best Paper Award 2023
Example Conference on Instrumentation (ECI 2023)

Certifications & Training

Deep Learning for Edge Devices Mar 2022
Example Academy of Applied Computing
Duration: 40 hours
Credential: EAAC-DL-0042
Valid until: Mar 2029

Technical Skills

Avionics	MIL-STD-1553 / ARINC 429, Avionics integration
Scientific instrumentation	Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems
Embedded systems	Embedded C, ESP32 / STM32 microcontrollers
Computer vision	Image super-resolution, Object detection
Drones & UAV systems	UAV payload integration
Edge AI	NVIDIA Jetson deployment

Languages

English, Hindi

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